BPAC — Literature Review Search Guide

Paper: *BPAC: How Detectable Is Regional Accent in Read Bangla? A Parallel Corpus and Its Phonetic Structure*

> **Framing principle for the whole review.** Your job is to make *"is regional accent detectable in read Bangla?"* read as a genuinely **open** question — so your subtle/null findings become the informative answer, not a disappointment. Every subsection should push toward: the question is open, the parallel corpus is the instrument that answers it, and the answer (subtle, uneven, organized by consonant timing) is the contribution.

> **Status:** §2.1 (Bangla speech corpora) — ✅ done.

How to use this document

1. Work the **starred subsections first** (§2.3 and the SSL half of §2.4) — they carry the framing.
2. Per search: grab the **2–3 most-cited** results + **1 recent** (last 2 years). Classics establish method; recent ones show currency.
3. When a paper is strongly relevant, follow Scholar's **"Cited by"** and **"Related articles"** rather than reformulating keywords — it snowballs faster.
4. Tag every entry with one of: `foundational-cite` (gets a real citation as method basis) · `context-cite` (supports a specific claim) · `skim-only` (grouped into a "several studies have…" sentence).
5. Anchor names to search directly: **Nerbonne** and **Heeringa** (dialectometry / acoustic-distance methods), and the **SUST Bangla corpus lineage** authors you already cite (their reference lists point to most existing Bangla-side literature).

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§2.2 — Parallel / accent-controlled corpus design

Job: justify *why parallelism matters*, with precedent — establish that a fixed script is a recognized tool for isolating the accent variable, so your design reads as principled rather than idiosyncratic. *(Some may overlap with §2.1 — use these to fill the design-justification angle.)*

Search terms

- `parallel speech corpus accent · parallel corpus dialect variation`
- `L2-ARCTIC accented speech corpus · ARCTIC speech database`
- `VCTK corpus multi-speaker accent`
- `read speech corpus phonetic coverage design`
- `sentence-controlled speech corpus dialect`
- `accented speech database design methodology`

Extract from each

1. Title / DOI / Year / Citations
2. Was it **parallel** (fixed script) or not?
3. What the parallel/controlled design **enabled** (the capability you're borrowing the justification from)
4. Language and domain
5. Tag: `foundational-cite / context-cite / skim-only`
6. Why parallelislm is necessary?

⭐ §2.3 — Regional / dialectal variation in Bangla *(the crucial subsection)*

Job: make the question open. Show prior Bangla dialect work is **mostly descriptive / text-based**, rarely quantitative-acoustic, and almost never on parallel data — so your measurements are new. Also introduce the **/s/→/ʃ/ observation as a hypothesis** (not established fact), so its near-null result in §7.5 reads as "tested a literature hypothesis, didn't hold at scale."

Search terms

- `Bangla dialect phonology · Bengali regional dialects`
- `Bengali dialectology · Bangladesh regional dialect variation`

- `Sylheti phonology · Sylheti Bengali dialect` *(directly relevant to your Sylhet findings)*
- `Chittagonian dialect Bengali · Chattogram dialect phonetics`
- `Bengali sibilant /ʃ/ /s/ · Bengali fricative realization dialect` *(sibilant-hypothesis provenance — important)*
- `Bengali vowel system acoustic · Bengali vowel inventory phonetics`
- `Bengali phonetics acoustic analysis`
- `Rangpur Rajshahi dialect Bengali` *(northern varieties, if anything exists)*

Extract from each

1. Title / DOI / Year / Citations
2. **Which regional variety/varieties** (map onto your 8 regions where possible)
3. **Method type** — `descriptive/impressionistic` vs. `text-based/computational` vs. `quantitative acoustic-phonetic` *(the key field — you're proving acoustic work is rare)*
4. **Phonetic feature studied** — vowels / consonants / sibilants / tone-pitch / rhythm / duration
5. **Data used** — how many speakers, read or spontaneous, parallel or not?
6. **Key finding** (one sentence)
7. **Strong or subtle/qualitative?** *(supports "detectability is uncertain")*
8. **The /s/→/ʃ/ or Sylhet-sibilant claim** — if present, capture the exact claim and whether it was **measured** or **asserted**
9. Tag: `foundational-cite / context-cite / skim-only`

> **Realistic expectation:** Bangla acoustic-phonetic dialectology is thin. **Finding little is the point** — it justifies your paper. Note the gap explicitly in the prose.

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§2.4 — Accent / dialect detection: methods and their limits

Job: pre-justify your null neural result *and* your multi-method approach. Two traditions, treated as complementary independent windows — which sets up §7.8's neural-vs-phonetic comparison.

⭐ Group A — Neural / SSL embeddings *(sets up your null §6)*

Search terms

- `accent classification deep learning · dialect identification neural network`
- `WavLM speech representation · self-supervised speech embeddings`
- `wav2vec 2.0 accent · speech foundation model dialect`
- `x-vector dialect identification · ECAPA-TDNN speaker embedding`
- `accent recognition low-resource language`
- `language variety identification embeddings limitations` *(look for papers reporting embeddings struggling on close varieties — **gold** for your framing)*
- `leave-one-speaker-out accent classification`

Extract from each

1. Title / DOI / Year / Citations
2. **Method** — WavLM / wav2vec2 / x-vector / ECAPA / other
3. **Task** — accent classification / dialect ID / language variety ID
4. **Varieties close or distant?** — dialects/accents of one language vs. different languages *(critical)*
5. **Reported accuracy / separability** — and against what chance baseline
6. **Did embeddings separate close varieties well or poorly?** *(most important field — any "struggles on close varieties" result is gold; capture it exactly)*
7. **Evaluation protocol** — speaker-independent (LOSO-style) or not?
8. Tag: `foundational-cite / context-cite / skim-only`

Group B — Classical acoustic-phonetic dialectometry *(sets up your §7)*

Search terms

- `dialectometry acoustic distance · dialect distance measurement phonetic`
- `Levenshtein dialect distance` *(Nerbonne / Heeringa — foundational)*
- `vowel formant dialect comparison · Lobanov normalization vowel dialect`
- `voice onset time dialect variation · VOT cross-dialect`
- `speech rhythm metrics dialect · nPVI rhythm typology language`
- `Pillai score vowel merger · vowel space area dialect`
- `spectral center of gravity sibilant · sibilant spectral moments dialect`
- `hierarchical clustering dialect dendrogram`

Extract from each

1. Title / DOI / Year / Citations
2. **Which measure** — vowel formants / VOT / rhythm metrics / Pillai / sibilant moments / distance-matrix-dendrogram
3. **Normalization used** — Lobanov or other? *(you use Lobanov — cite precedent)*
4. **How the distance/clustering was built** *(the template you follow in §7.7)*
5. **Did the measure discriminate dialects well?** — effect sizes if given
6. **Foundational vs. applied** — method-defining (cite as basis) or application (cite as precedent)
7. Tag: `foundational-cite / context-cite / skim-only`

> **Method papers to secure as `foundational-cite`:** Lobanov (vowel normalization), Grabe & Low / Ramus (rhythm metrics — nPVI etc.), Nerbonne & Heeringa (dialectometry / acoustic distance). Cite recent SSL papers for currency.

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§2.5 — Read vs. spontaneous speech & detectability

Job: the direct on-ramp to your title. Establish that read speech is more controlled but tends to **attenuate** accent features — so detectability in *read* Bangla is genuinely uncertain and worth measuring. End the review here by stating the open question and the gap you fill.

Search terms

- `read versus spontaneous speech acoustic differences`
- `speaking style accent variation · read speech reduces coarticulation`
- `elicited speech dialect features attenuation`
- `speech style effect on phonetic variation`

Extract from each

1. Title / DOI / Year / Citations
2. **What was compared** — read vs. spontaneous, or across styles
3. **Direction of effect** — does read speech *attenuate* accent/coarticulation/variation? *(you want: read reduces natural variation)*
4. **Which features affected** — vowels / reduction / rhythm
5. **One-sentence takeaway**

6. Tag: `foundational-cite` / `context-cite` / `skim-only`

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§2.6 Applications requiring parallel speech data —

- `parallel data voice conversion` · `non-parallel voice conversion` *(the contrast is what makes the point)*
- `accent conversion parallel corpus` · `foreign accent conversion training data`
- `sequence-to-sequence voice conversion parallel` · `frame-level alignment voice conversion`
- `dynamic time warping voice conversion training`
- `speech-to-speech translation parallel corpus`
- `accent conversion evaluation reference speech`

Extract from each paper

1. **Title / DOI / Year / Citations**

2. **Task** — voice conversion / accent conversion / speech-to-speech translation / expressive conversion / other. *(Keeps the cluster organized; accent conversion is closest to your case.)*

3. **Parallel data: required, beneficial, or not used?** — the central field. Tag one of:

   - `requires-parallel` (architecture cannot train without aligned pairs)
   - `benefits-from-parallel` (works either way, better with)
   - `non-parallel` (explicitly avoids the need)

   You need examples of all three. The `requires-parallel` ones justify your corpus directly; the `non-parallel` ones are what you must acknowledge (most modern systems avoid parallel data — your §2.6 shouldn't overclaim by hiding that).

4. **If parallel: what alignment method?** — DTW frame alignment, forced-alignment-based, attention/seq2seq implicit alignment. *(Tells you which methods your corpus could feed, and connects to your own MFA infrastructure.)*

5. **Stated data bottleneck** — does the paper say parallel data is **scarce, expensive, or unavailable**, especially for low-resource languages? **Capture the sentence verbatim if so.** *(This is your gold quote — it's the single most useful thing in this cluster, because it turns "we built a parallel corpus" into "we built the thing the literature says is missing.")*

6. **Did they use parallel data for evaluation even when training non-parallel?** — many systems train without parallel data but *evaluate* against same-content references for objective metrics (MCD, content preservation). *(Critical for your §8 claim: it means your corpus is useful even to non-parallel systems, which is a much broader claim than "some architectures need it.")*

7. **Language(s) and data scale** — how many speakers/hours of parallel data did they have? *(Lets you position BPAC's 27.7 h / 51 speakers / 250 parallel sentences against precedent — and whether anything comparable exists for a South Asian language.)*

8. **Accent conversion specifically: what was the source–target pairing?** — L2→L1, regional→standard, speaker→speaker? *(Your corpus is regional→standard, which is less common than L2→L1; if precedent is thin, that's another gap you fill.)*

9. **Evaluation metrics used** — MCD, WER, MOS, accent-ID accuracy, ECAPA similarity. *(Feeds your §8 evaluation-protocol argument: which metrics require a same-content reference.)*

10. **Tag:** `foundational-cite / context-cite / skim-only`